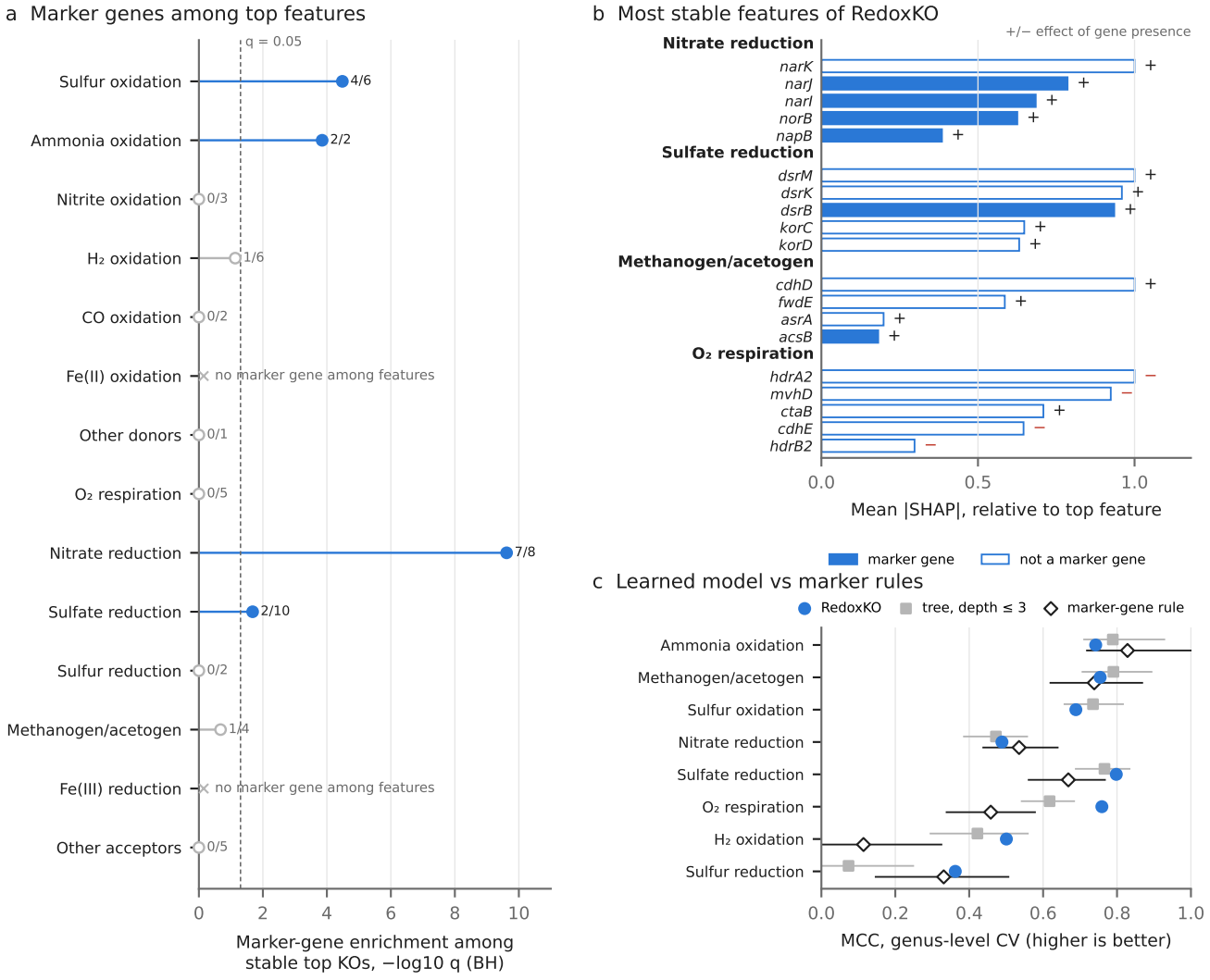
a Marker genes among top features
q = 0.05
Sulfur oxidation
4/6
Ammonia oxidation
2/2
Nitrite oxidation
0/3
H₂ oxidation
1/6
CO oxidation
0/2
Fe(II) oxidation
no marker gene among features
Other donors
0/1
O₂ respiration
0/5
Nitrate reduction
7/8
Sulfate reduction
2/10
Sulfur reduction
0/2
Methanogen/acetogen
1/4
Fe(III) reduction
no marker gene among features
Other acceptors
0/5
0
2
4
6
8
10
Marker-gene enrichment among stable top KOs, −log10 q (BH)
b Most stable features of RedoxKO
+/− effect of gene presence
Nitrate reduction
narK
narJ
narI
norB
napB
Sulfate reduction
dsrM
dsrK
dsrB
korC
korD
Methanogen/acetogen
cdhD
fwdE
asrA
acsB
O₂ respiration
hdrA2
mvhD
ctaB
cdhE
hdrB2
0.0
0.5
1.0
Mean |SHAP|, relative to top feature
marker gene
not a marker gene
c Learned model vs marker rules
RedoxKO
tree, depth ≤ 3
marker-gene rule
Ammonia oxidation
Methanogen/acetogen
Sulfur oxidation
Nitrate reduction
Sulfate reduction
O₂ respiration
H₂ oxidation
Sulfur reduction
0.0
0.2
0.4
0.6
0.8
1.0
MCC, genus-level CV (higher is better)